

Euclid's Elements — Book V

Proposition 15

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Parts have the same ratio as their equimultiples.

1 Introduction

Proposition V.15 records the invariance of a ratio under taking the same positive multiple of both terms.

In the present reconstruction the proposition is formulated for the positive multiple operation already defined in the Book V interface. If $M_k(a)$ denotes the $(k + 1)$ -fold positive multiple of a , then the theorem states

$$a : b = M_k(a) : M_k(b).$$

The proof is direct from Eudoxus' Definition 5. No Archimedean argument is needed, and no strict greater-ratio witness is constructed.

The production file is

```
CGJteamLab/Proposition5_15.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the shared Book V magnitude infrastructure.

2 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_15
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b : HilbertPositiveSegmentClass Geo)
  (k : Nat) :
  HilbertEudoxusProportion
    Geo
    a b
    (hilbertPositiveSegmentMultiple Geo k a)
    (hilbertPositiveSegmentMultiple Geo k b) := by

intro m n

constructor
case left =>
  exact
    (hilbertPositiveSegmentMultiple_nested_lt_iff
      Geo m n k a b).symm

case right =>
  constructor
  case left =>
    exact
      (hilbertPositiveSegmentMultiple_nested_eq_iff
        Geo m n k a b).symm
  case right =>
    exact
      (hilbertPositiveSegmentMultiple_nested_lt_iff
        Geo n m k b a).symm

```

The code shown here is an ASCII presentation of the same proof structure as the production theorem.

3 Positive Multiples

The operation

```

hilbertPositiveSegmentMultiple Geo k a

```

uses shifted indexing:

$$M_0(a) = a, \quad M_1(a) = a + a, \quad M_2(a) = a + a + a, \quad \dots$$

Thus every natural-number index represents a positive multiple.

For Proposition V.15, the same fixed index k is applied to both terms of the ratio.

4 What Definition 5 Requires

To prove

$$a : b = M_k(a) : M_k(b),$$

Definition 5 requires that for every pair of positive-multiple indices m, n , the following three comparisons agree:

$$M_m(a) < M_n(b),$$

$$M_m(a) = M_n(b),$$

and

$$M_n(b) < M_m(a),$$

with the corresponding comparisons after applying the common outer multiple k .

Thus the formal target reduces to three nested-multiple equivalences.

5 Nested Multiples

The reusable infrastructure proves:

```
hilbertPositiveSegmentMultiple_nested_lt_iff
hilbertPositiveSegmentMultiple_nested_eq_iff
```

Conceptually these state that applying the same positive outer multiple to both sides preserves and reflects strict order and equality.

For example,

$$M_m(M_k(a)) < M_n(M_k(b)) \iff M_m(a) < M_n(b),$$

and similarly for equality.

The reverse strict comparison is obtained by applying the same theorem with the two magnitudes and indices exchanged.

6 Why No Arithmetic of Scalar Multiplication Is Needed

One might expect a proof based on the numerical identity

$$(m + 1)(k + 1) = (k + 1)(m + 1).$$

The formal development avoids introducing scalar multiplication of magnitudes by natural numbers as a separate algebraic structure.

Instead, the required commutation of iterated positive addition is already isolated in the interface theorem

```
hilbertPositiveSegmentMultiple_commute
```

and the nested order/equality lemmas are built on top of it.

Thus V.15 remains internal to the synthetic additive magnitude layer.

7 Lean Proof Architecture

The complete proof has the form

$$\begin{array}{c}
\text{fix arbitrary } m, n \\
\Downarrow \\
M_m(a) < M_n(b) \iff M_m(M_k(a)) < M_n(M_k(b)) \\
\Downarrow \\
M_m(a) = M_n(b) \iff M_m(M_k(a)) = M_n(M_k(b)) \\
\Downarrow \\
M_n(b) < M_m(a) \iff M_n(M_k(b)) < M_m(M_k(a)) \\
\Downarrow \\
a : b = M_k(a) : M_k(b).
\end{array}$$

There are no proposition-local helper lemmas.

8 Axiomatic Strength

The proposition requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

through the positive segment-class infrastructure.

In particular, it does not require

```
HilbertArchimedeanPlane
```

and therefore does not depend on the Archimedean step used in V.8.

This is structurally natural: V.15 does not assert the existence of a sufficiently large multiple. It only compares already specified positive multiples.

9 Relationship with Definition 5

V.15 is almost a normalization theorem for Definition 5.

The equality of ratios is invariant under applying the same positive multiple to both terms because all three comparison types are invariant under that operation.

The proposition therefore demonstrates that the Eudoxean definition is compatible with the additive multiple structure developed earlier in `HilbertInterfaceV.lean`.

10 Relationship with V.16

V.15 is used twice in V.16.

Fix arbitrary indices m, n . From V.15 we get

$$a : b = M_m(a) : M_m(b)$$

and

$$c : d = M_n(c) : M_n(d).$$

Starting from

$$a : b = c : d,$$

V.11 composes these equalities to obtain

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

Then V.14 transfers the comparison of the first terms to the second terms.

Thus the proposition-level dependency chain is

$$\text{V.15} \longrightarrow \text{V.11} \longrightarrow \text{V.14} \longrightarrow \text{V.16}.$$

11 Hidden Formal Data

The main formal datum hidden in the classical statement is the indexing convention.

Because the project uses positive multiples only, the natural index k denotes $k + 1$ copies. Consequently the theorem is valid uniformly for all

$$k : \mathbb{N},$$

including $k = 0$, where it becomes the reflexive proportion

$$a : b = a : b.$$

No separate positivity proof for the multiplier is required.

12 Classical DAG versus Formal DAG

At the mathematical level:

$$\begin{array}{c}
 a : b \\
 \Downarrow \text{common positive multiple } k \\
 M_k(a) : M_k(b) \\
 \Downarrow \\
 a : b = M_k(a) : M_k(b).
 \end{array}$$

At the Lean level:

$$\begin{array}{c}
 \text{nested_lt_iff} \\
 \text{nested_eq_iff} \\
 \text{nested_lt_iff with reversed arguments} \\
 \Downarrow \\
 \text{three clauses of Definition 5} \\
 \Downarrow \\
 \text{euclid_proposition_5_15.}
 \end{array}$$

The formal proof is therefore a direct structural reading of the definition.

13 Structural Lesson

V.15 shows that once positive multiples have been given a clean reusable interface, a classical proposition about equimultiples can become very short.

The mathematical work lies below the Euclid-numbered theorem: commutation of iterated addition and preservation/reflection of order and equality under a common positive multiple.

Keeping those facts in `HilbertInterfaceV.lean` leaves `Proposition5_15.lean` as the correct proposition-level statement rather than as a second arithmetic library.

14 Conclusion

Proposition V.15 establishes

$$a : b = M_k(a) : M_k(b)$$

for every positive-multiple index k .

Its proof is a direct verification of Eudoxus' Definition 5 using the nested multiple order and equality equivalences already available in the shared Book V interface.

The proposition is formally small but strategically important: it supplies the scaling step used twice in the proof of V.16 alternando.